UNIVERSITY OF BIRMINGHAM

School of Physics and Astronomy

DEGREE OF M.Sci. WITH HONOURS

FOURTH-YEAR FINAL EXAMINATION

03 01087

LM NUCLEAR PHYSICS

SEMESTER 2 EXAMINATIONS 2021/22

Time Allowed: 1 hour 30 minutes

Answer one question from Section 1 and one question from Section 2.

Section 1 consists of two questions and carries 50% of the marks for the examination. Answer ***one*** question from this Section. If you answer more than one question, credit will only be given for the best answer.
Section 2 consists of a single question and carries 50% of the marks for the examination.

The approximate allocation of marks to each part of a question is shown in brackets **[ ]**.

All symbols have their usual meanings.

Calculators may be used in this examination but must not be used to store text. Calculators with the ability to store text should have their memories deleted prior to the start of the examination.

A formula sheet and a table of physical constants and units that may be required will be found at the end of this question paper.

SECTION 1

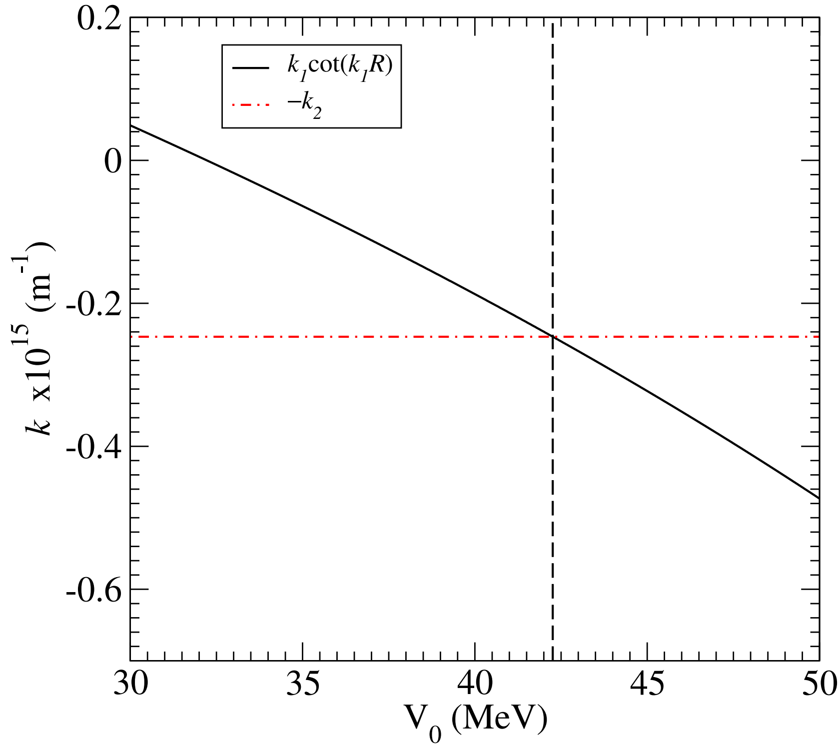
Answer **one** *question from this Section. If you answer more than one question, credit will only be given for the best answer.*

1. (a) Consider the properties of the strong force and explain why the deuteron is bound but the di-neutron and di-proton are not. **[5]**

(b) Referring to measured properties of the deuteron ground state, fully explain the allowed orbital angular momentum quantum numbers. **[5]**

(c) The measured magnetic moment of the deuteron is 0.8574376 μ_N. Calculate the relative magnitude of each orbital angular momentum contribution. [For the neutron $g_s = -3.826084$ and for the proton $g_s = +5.585691$.] **[8]**

(d) Use the plot below to estimate the value of the deuteron binding energy and the radius.



[6]

(e) The analysis that led to the plot of part (d) relies on the assumption of a central potential. Discuss the measured properties of the deuteron that are inconsistent with this assumption and the implications that follow. **[6]**

2. (a) For electromagnetic transitions

$$\lambda(L) = \frac{2(L+1)}{\varepsilon_0 \hbar L[(2L+1)!!]^2} \left(\frac{\omega}{c}\right)^{2L+1} [B(L)],$$

where the symbols have their usual meaning.

i. Show for a radial operator given by $\hat{O} = er^L$ that for electric transitions the reduced transition probability, $B(EL)$, is given by

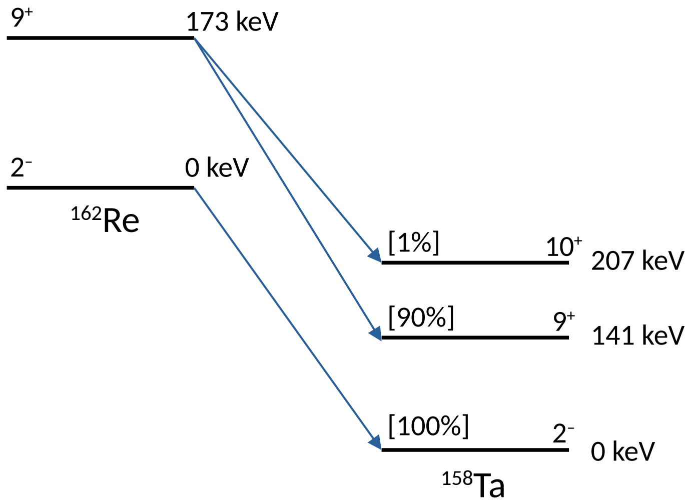
$$B(EL) = e^2 \left(\frac{3R^L}{L+3}\right)^2,$$

where R is the nuclear radius. State any assumptions you make. **[5]**

ii. What is the energy and mass dependence for electric dipole radiation? **[3]**

Question continues on next page.

(b) The diagram shows the decay of the ground state and first excited state of $^{162}_{75}$Re to $^{158}_{73}$Ta. The half-lives of the two states are similar with 107 ms and 77 ms, respectively. The mass excesses are $\Delta m(^{162}\mathrm{Re}) = -22453.0$ keV, $\Delta m(^{158}\mathrm{Ta}) = -31118.0$ keV and $\Delta m(^{4}\mathrm{He}) = 2424.9$ keV.



i. Calculate the energies of the three alpha particles for the decay branches shown in the diagram. **[3]**

ii. Explain the alpha-particle branching ratios and state the orbital angular momentum of the alpha particles. Why does the 173 keV level in ^{162}Re preferentially alpha decay rather than gamma decay? **[4]**

iii. State the multipolarities of the gamma-ray transitions between the 207 keV and 141 keV states and the 141 keV level and the ground state in ^{158}Ta. **[3]**

iv. One hundred thousand decays of the 141 keV level in ^{158}Ta are monitored, but no gamma rays are observed. Show quantitatively why and write down the preferred decay mode. **[4]**

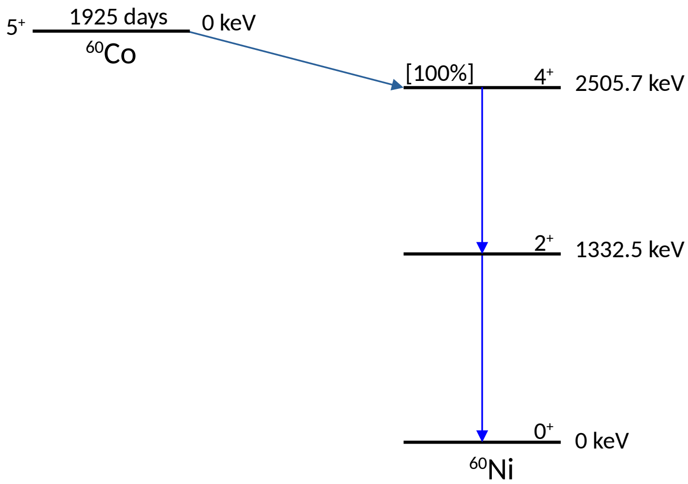
v. The excited state of ^{162}Re can also proton decay to the $^{161}_{74}$W ground state. The mass excesses are $\Delta m(^{161}\mathrm{W}) = -30507.0$ keV and $\Delta m(^{1}\mathrm{H}) = 7289.0$ keV. Calculate the barrier, B, Q-value and show whether the Gamow factor, G is larger or smaller than for the alpha-decay to the 141 keV state in ^{158}Ta. Comment on your result. **[8]**

SECTION 2

Answer **one** *question from this Section.*

3. (a) In the doubly magic nucleus $^{208}_{82}$Pb, the last filled neutron shell is $3p_{1/2}$ and the first empty neutron shell is $2g_{9/2}$. In spectroscopic notation g corresponds to $l = 4$. The three lowest lying states are
$E_x = 2.615$ MeV ($I^\pi = 3^-$),
$E_x = 3.198$ MeV ($I^\pi = 5^-$) and
$E_x = 3.475$ MeV ($I^\pi = 4^-$).
In the $j - j$ coupling scheme, which state cannot be described simply in the shell model? Explain the configurations of all three states in detail. **[8]**

(b) The standard calibration γ-source $^{60}_{27}$Co is based on the scheme given in the figure below.



i. Explain why ^{60}Co does not decay directly into the ground state of the $^{60}_{28}$Ni nucleus. **[4]**

ii. Show quantitatively why two photons are emitted in the subsequent de-excitation of the ^{60}Ni nucleus, rather than just one. **[4]**

iii. Estimate the lifetime of the 4^+ state in ^{60}Ni and comment on the accuracy of your result. **[2]**

Question continues on next page.

(c) A rotational band based on the ground-state configuration of $^{20}_{10}$Ne contains the excited states at 1.63 MeV ($I^\pi = 2^+$), 4.25 MeV ($I^\pi = 4^+$) and 8.76 MeV ($I^\pi = 6^+$).
[$\Delta m(^{20}\mathrm{Ne}) = -7041.93213$ keV, $\Delta m(^{19}\mathrm{Ne}) = 1752.05396$ keV, $\Delta m(^{19}\mathrm{F}) = -1487.44507$ keV, $\Delta m(^{16}\mathrm{O}) = -4737.00195$ keV, $\Delta m(^{4}\mathrm{He}) = 2424.91577$ keV, $\Delta m(^{1}\mathrm{H}) = 7288.97119$ keV and $\Delta m(^{1}n) = 8071.31787$ keV.]

i. Calculate the average moment of inertia for the rotational band and specify in standard units. **[4]**

ii. Give a reaction suitable for populating the $I^\pi = 6^+$ state and describe how each of the levels in the band are likely to decay. **[5]**

iii. Explain why states in the band with spins higher than $I = 8$ are not expected. **[3]**

Useful Formulae for Nuclear Physics

Coulomb barrier, $V_{Coul}(r) = \dfrac{Z_1 Z_2 e^2}{4\pi\varepsilon_0 R}$, where $R = 1.2A^{1/3}$ fm

Radial Schrödinger equation, $-\dfrac{\hbar^2}{2m}\dfrac{d^2u}{dr^2} + \left[V(r) + \dfrac{l(l+1)\hbar^2}{2mr^2}\right]u(r) = Eu(r)$,

where $R(r) = \dfrac{u(r)}{r}$

Deuteron wavefunction matching condition, $k_1 \cot(k_1 R) = -k_2$

where $k_1 = \sqrt{\dfrac{2m(E+V_0)}{\hbar^2}}$ and $k_2 = \sqrt{-\dfrac{2mE}{\hbar^2}}$

Magnetic moment, $\langle\mu\rangle =$

$$\frac{1}{2(j+1)}\left\{g_l\left[j(j+1)+l(l+1)-s(s+1)\right]+g_s\left[j(j+1)+s(s+1)-l(l+1)\right]\right\}\mu_N$$

Alpha-decay half-life, $t_{1/2} = \ln(2)\dfrac{R}{c}\sqrt{\dfrac{mc^2}{2(V_0+Q)}}\exp\{2G\}$,

where G is the Gamow factor, $G = \sqrt{\dfrac{2m}{\hbar^2 Q}}\dfrac{Z_1 Z_2 e^2}{4\pi\varepsilon_0}\left[\dfrac{\pi}{2} - 2\left(\dfrac{Q}{B}\right)^{1/2}\right]$

Beta-decay ft values, $f(Z,T_e)t_{1/2} = \ln(2)\dfrac{2\pi^3\hbar^7}{g_F^2|M_{fi}|^2 m_e^5 c^4}$

Internal conversion coefficients, $\alpha(ML) \approx \dfrac{Z^3}{n^3}\left(\dfrac{e^2}{4\pi\varepsilon_0\hbar c}\right)^4\left(\dfrac{2m_ec^2}{E}\right)^{L+3/2}$ and

$$\alpha(EL) \approx \frac{Z^3}{n^3}\left(\frac{L}{L+1}\right)\left(\frac{e^2}{4\pi\varepsilon_0\hbar c}\right)^4\left(\frac{2m_ec^2}{E}\right)^{L+5/2}$$

Weisskopf estimates (for E_γ in MeV),

$\lambda(E1) = 1.0\times10^{14}A^{2/3}E_\gamma^3$

$\lambda(M1) = 3.1\times10^{13}E_\gamma^3$

$\lambda(E2) = 7.4\times10^{7}A^{4/3}E_\gamma^5$

$\lambda(M2) = 2.2\times10^{7}A^{2/3}E_\gamma^5$

$\lambda(E3) = 3.5\times10^{1}A^{2}E_\gamma^7$

$\lambda(M3) = 1.1\times10^{1}A^{4/3}E_\gamma^7$

$\lambda(E4) = 1.1\times10^{-5}A^{8/3}E_\gamma^9$

$\lambda(M4) = 3.3\times10^{-6}A^{2}E_\gamma^9$

$\lambda(E5) = 2.4\times10^{-12}A^{10/3}E_\gamma^{11}$

$\lambda(M5) = 7.4\times10^{-13}A^{8/3}E_\gamma^{11}$

Physical Constants and Units

Acceleration due to gravity	g	9.81 m s^{-2}
Gravitational constant	G	6.674×10^{-11} N m^2 kg^{-2}
Ice point	T_{ice}	273.15 K
Avogadro constant	N_A	6.022×10^{23} mol^{-1}
		[*N.B.* 1 mole $\equiv 1$ *gram-molecule*]
Gas constant	R	8.314 J K^{-1} mol^{-1}
Boltzmann constant	k, k_B	1.381×10^{-23} J K^{-1} $\equiv 8.62 \times 10^{-5}$ eV K^{-1}
Stefan constant	σ	5.670×10^{-8} W m^{-2} K^{-4}
Rydberg constant	R_∞	1.097×10^{7} m^{-1}
	$R_\infty hc$	13.606 eV
Planck constant	h	6.626×10^{-34} J s $\equiv 4.136 \times 10^{-15}$ eV s
$h/2\pi$	$\hbar$	1.055×10^{-34} J s $\equiv 6.582 \times 10^{-16}$ eV s
Speed of light *in vacuo*	c	2.998×10^{8} m s^{-1}
	$\hbar c$	197.3 MeV fm
Charge of proton	e	1.602×10^{-19} C
Mass of electron	m_e	9.109×10^{-31} kg
Rest energy of electron		0.511 MeV
Mass of proton	m_p	1.673×10^{-27} kg
Rest energy of proton		938.3 MeV
One atomic mass unit	u	1.66×10^{-27} kg
Atomic mass unit energy equivalent		931.5 MeV
Electric constant	ϵ_0	8.854×10^{-12} F m^{-1}
Magnetic constant	μ_0	$4\pi \times 10^{-7}$ H m^{-1}
Bohr magneton	μ_B	9.274×10^{-24} A m^2 (J T^{-1})
Nuclear magneton	μ_N	5.051×10^{-27} A m^2 (J T^{-1})
Fine-structure constant	$\alpha = e^2/4\pi\epsilon_0\hbar c$	7.297×10^{-3}= 1/137.0
Compton wavelength of electron	$\lambda_c = h/m_e c$	2.426×10^{-12} m
Bohr radius	a_0	5.2918×10^{-11} m
angstrom	Å	10^{-10} m
barn	b	10^{-28} m^2
torr (mm Hg at 0 °C)	torr	133.32 Pa (N m^{-2})

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<u>Do not</u> complete the attendance slip, fill in the front of the answer book or turn over the question paper until you are told to do so.

Important Reminders

- Coats/outwear should be placed in the designated area.
- Unauthorised materials (e.g. notes or Tippex) MUST be placed in the designated area.
- Check that you DO NOT have any unauthorised materials with you (e.g. in your pockets, pencil case).
- Mobile phones and smart watches MUST be switched off and placed in the designated area or under your desk. They must not be left on your person or in your pockets.
- You are NOT permitted to use a mobile phone as a clock. If you have difficulty seeing a clock, please alert an Invigilator.
- You are NOT permitted to have writing on your hand, arm or other body part.
- Check that you do not have writing on your hand, arm or other body part – if you do, you must inform an Invigilator immediately.
- Alert an Invigilator immediately if you find any unauthorised item upon you during the examination.

Any students found with non-permitted items upon their person during the examination, or who fail to comply with Examination rules may be subject to Student Conduct Procedures